RACE
Regular Analysis Through Continuous Exercise

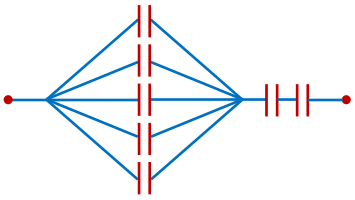
Capacitance : RACE-01

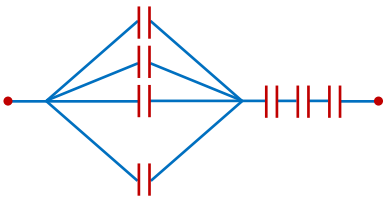
1. A parallel plate capacitor has $d = 1mm$ and $C = 1\ F$ with no medium inside will have:-

(A) Area = $36\ \pi \times 10^6$ (B) Area = $4\ \pi \times 10^6$ (C) Area = $6\ \pi \times 10^6$ (D) Area = $\pi \times 10^6$

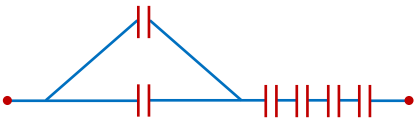
PCC150

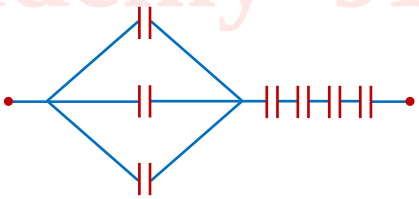
2. Seven capacitors, each of capacitance 2 μF, are to be connected in a configuration to obtain an effective capacitance of $\left(\frac{6}{13}\right)\mu F$. Which of the combinations, shown in figures below, will achieve the desired value ?

(A) 

(B) 

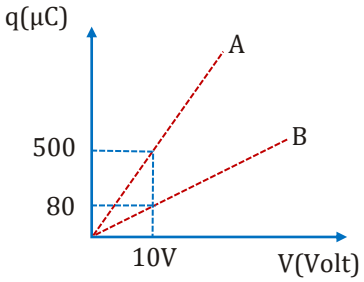
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(C) 

(D) 

PCC151

3. Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. The capacitances are :

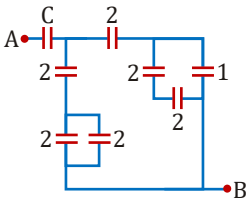


(A) 50 μF and 30 μF (B) 20 μF and 30 μF (C) 60 μF and 40 μF (D) 40 μF and 10 μF

PCC152

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4. In the circuit shown, find C if the effective capacitance of the whole circuit is to be 0.5 μF. All values in the circuit are in μF.

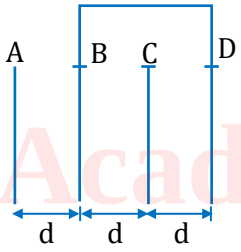


(A) $\frac{7}{10}\mu F$ (B) $\frac{7}{11}\mu F$ (C) $\frac{6}{5}\mu F$ (D) $4\mu F$

PCC153

5. Four identical rectangular plates with length, $l = 2\ cm$ and breadth, $b = \frac{3}{2}\ cm$ are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\varepsilon_0}{d}$. The value of x is _____.

(Round off to the Nearest Integer)



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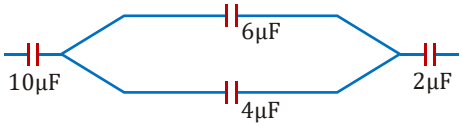
PCC154

6. Capacitance of an isolated conducting sphere of radius R_1 becomes n times when it is enclosed by a concentric conducting sphere of radius R_2 connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is:

(A) $\frac{n}{n-1}$ (B) $\frac{2n}{2n+1}$ (C) $\frac{n+1}{n}$ (D) $\frac{2n+1}{n}$

PCC155

7. In the figure shown below, the charge on the left plate of the 10 μF capacitor is –30 μC ? The charge on the right plate of the 6 μF capacitor is :

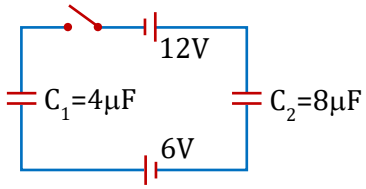


(A) –18 μC (B) –12 μC (C) +12 μC (D) +18 μC

PCC156

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8. In the circuit shown initially C_1 and C_2 are uncharged. After closing the switch:



(A) The charge on C_2 is greater that on C_1.
(B) The charge on C_1 and C_2 are the same.
(C) The potential drops across C_1 and C_2 are the same.
(D) The potential drops across C_2 is greater than that across C_1.

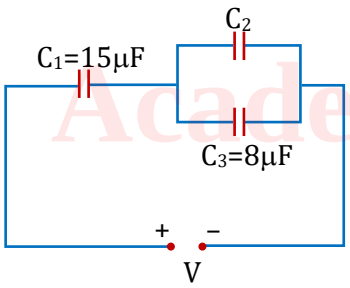
PCC157

9. A 10 μF capacitor is fully charged to a potential difference of 50V. After removing the source voltage, it is connected to an uncharged capacitor in parallel. Now the potential difference across them becomes 20V. The capacitance of the second capacitor is:

(A) 10 μF (B) 15 μF (C) 20 μF (D) 30 μF

PCC158

10. In the circuit shown in the figure, the total charge in 750 μC and the voltage across capacitor C_2 is 20V. Then the charge on capacitor C_2 is:



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(A) 590 μC (B) 450 μC (C) 650 μC (D) 160 μC

PCC159

ANSWER KEY

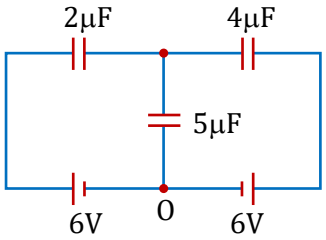
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	D	B	2	A	D	B	B	A

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Capacitance : RACE-02

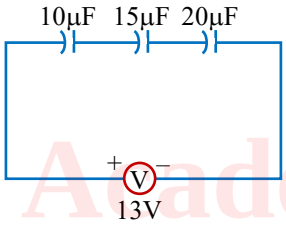
1. In the circuit shown, charge on the 5 μF capacitor is :



(A) 5.45 μC (B) 16.36 μC (C) 10.90 μC (D) 18.00 μC

PCC160

2. The charge on capacitor of capacitance 15μF in the figure given below is :



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(A) 60 μC (B) 130 μC (C) 260 μC (D) 585 μC

PCC161

3. A force of 10N acts on a charged particle placed between two plates of a charged capacitor. If one plate of capacitor is removed, then the force acting on that particle will be :

(A) 5 N (B) 10 N (C) 20 N (D) Zero

PCC162

4. Two capacitors of capacitances C and $2C$ are charged to potential differences V and $2V$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is:

(A) $\frac{9}{2}CV^2$ (B) $\frac{25}{6}CV^2$ (C) Zero (D) $\frac{3}{2}CV^2$

PCC163

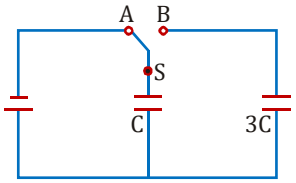
5. If the charge on a capacitor is increased by $2C$, the energy stored in it increases by 44%. The original charge on the capacitor is (in C) :

(A) 10 (B) 20 (C) 30 (D) 40

PCC164

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6. In the figure shown, after the switch 'S' is turned from position 'A' to position 'B', the energy dissipated in the circuit in terms of capacitance 'C' and total charge 'Q' is:



(A) $\frac{3}{8}\frac{Q^2}{C}$ (B) $\frac{3}{4}\frac{Q^2}{C}$ (C) $\frac{1}{8}\frac{Q^2}{C}$ (D) $\frac{5}{8}\frac{Q^2}{C}$

PCC165

7. Effective capacitance of parallel combination of two capacitors C_1 and C_2 is 10 μF. When these capacitors are individually connected to a voltage source of $1V$, the energy stored in the capacitor C_2 is 4 times that of C_1. If these capacitors are connected in series, their effective capacitance will be:

(A) 3.2 μF (B) 8.4 μF (C) 1.6 μF (D) 4.2 μF

PCC166

8. A 5 μF capacitor is charged fully by a 220 V supply. It is then disconnected from the supply and is connected in series to another uncharged 2.5 μF capacitor. If the energy change during the charge redistribution is $\frac{X}{100}J$ then value of X to the nearest integer is_____.

PCC167

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9. An electron with kinetic energy K_1 enters between parallel plates of a capacitor at an angle 'α' with the plates. It leaves the plates at angle 'β' with kinetic energy K_2. Then the ratio of kinetic energies $K_1 : K_2$ will be :

(A) $\frac{\sin^2\beta}{\cos^2\alpha}$ (B) $\frac{\cos^2\beta}{\cos^2\alpha}$ (C) $\frac{\cos\beta}{\cos\alpha}$ (D) $\frac{\cos\beta}{\sin\alpha}$

PCC168

ANSWER KEY

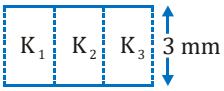
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Answer	B	A	A	D	A	A	C	4	B

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RACE Regular Analysis Through Continuous Exercise

Capacitance : RACE-03

1. A parallel plate capacitor is of area 6 cm^2 and a separation 3 mm. The gap is filled with three dielectric materials of equal thickness (see figure) with dielectric constants $K_1 = 10$, $K_2 = 12$ and $K_3 = 14$. The dielectric constant of a material which when fully inserted in above capacitor, gives same capacitance would be:



(A) 12 (B) 4 (C) 36 (D) 14

PCC169

2. Voltage rating of a parallel plate capacitor is 500V. Its dielectric can withstand a maximum electric field of 10^6 V/m. The plate area is 10^{-4} m^2. What is the dielectric constant if the capacitance is 15 pF? (Given $\in_0 = 8.86 \times 10^{-12}$ C^2/Nm^2)

(A) 3.8 (B) 4.5 (C) 6.2 (D) 8.5

PCC170

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3. A parallel plate capacitor has plate of length 'l', width 'w' and separation of plates is 'd'. It is connected to a battery of emf V. A dielectric slab of the same thickness 'd' and of dielectric constant $k = 4$ is being inserted between the plates of the capacitor. At what length of the slab inside plates, will be energy stored in the capacitor be two times the initial energy stored?

(A) $l/4$ (B) $l/2$ (C) $l/3$ (D) $2l/3$

PCC171

4. A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference $V = 12V$ between its plates. The charging battery is now disconnected and a porcelin plate with $k = 7$ is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ. (Assume no friction)

PCC172

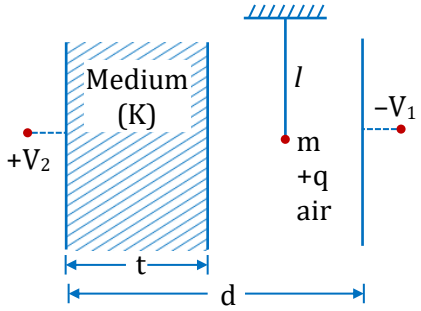
5. The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V. When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is :-

(A) $\frac{V}{K+n}$ (B) V (C) $\frac{(n+1)V}{(K+n)}$ (D) $\frac{nV}{K+n}$

PCC173

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6. A simple pendulum of mass 'm', length 'l' and charge '$+q$' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be:



(A) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_1(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

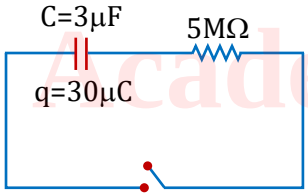
(B) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

(C) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_2(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

(D) $\tan^{-1}\left[\frac{q}{mg}\times\frac{C_1(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

PCC174

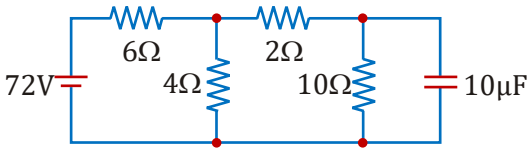
7. The circuit shown in the figure consists of a charged capacitor of capacity 3 μF and a charge of 30 μC. At time $t = 0$, when the key is closed, the value of current flowing through the 5 $M\Omega$ resistor is 'x' μ-A. The value of 'x to the nearest integer is _____.



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PCC175

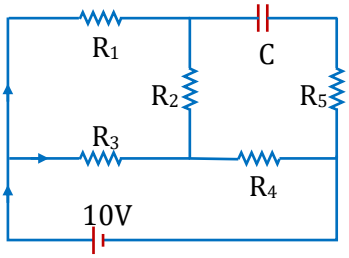
8. Determine the charge on the capacitor in the following circuit :



(A) 2 μC (B) 60 μC (C) 200 μC (D) 10 μC

PCC176

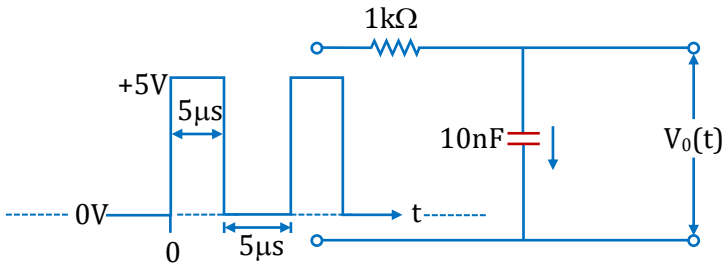
9. An ideal cell of emf 10 V is connected in circuit shown in figure. Each resistance is 2 Ω. The potential difference (in V) across the capacitor when it is fully charged is _____.

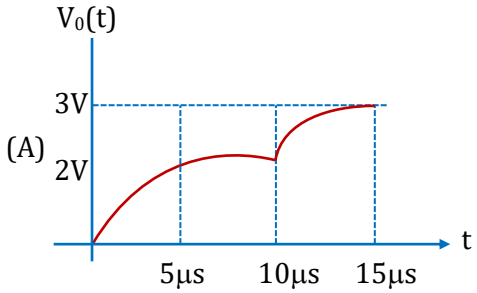


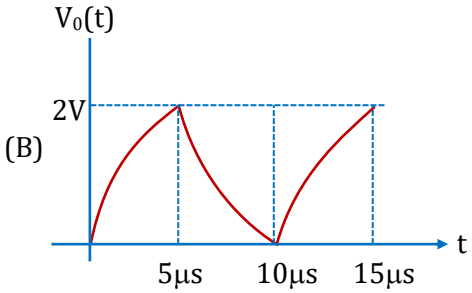
PCC177

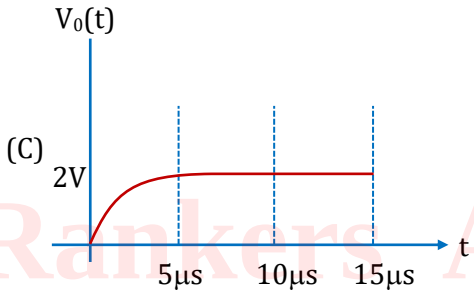
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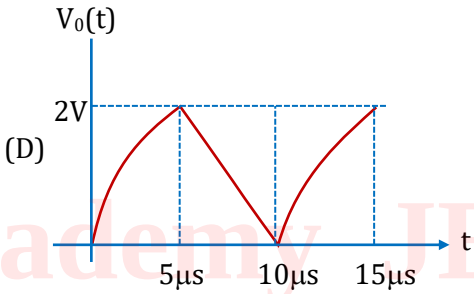
10. For the given input voltage waveform $V_{in}(t)$, the output voltage waveform $V_D(t)$, across the capacitor is correctly depicted by:











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PCC178

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	C	864	C	C	2	C	8	A

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